Homework sheet 1

Logic Programming (621.315), Winter semester 2026

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Exercise 1.1

Determine the answer set of the following positive logic programs.

(a)

Rule numbers refer to the following order (one rule ends at each period).

```
a :- b, d.
a :- c, e.
b :- d, m.
b :- e, f.
c :- d, e.
d.
e :- d.
f :- b, c.
g :- d, f.
g :- e, h.
h :- c, k.
k :- a, e.
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{d}\} \qquad \text{fact, rule 6}$$

$$M_2 = M_1 \cup \{\texttt{e}\} \qquad \text{rule 7}$$

$$M_3 = M_2 \cup \{\texttt{c}\} \qquad \text{rule 5}$$

$$M_4 = M_3 \cup \{\texttt{a}\} \qquad \text{rule 2}$$

$$M_5 = M_4 \cup \{\texttt{k}\} \qquad \text{rule 12}$$

$$M_6 = M_5 \cup \{\texttt{h}\} \qquad \text{rule 11}$$

$$M_7 = M_6 \cup \{\texttt{g}\} \qquad \text{rule 10}$$

No rule has a true body and a head outside M_7; hence this is the least fixed point.

$$\mathrm{Cn}(R) = \{\texttt{a}, \texttt{c}, \texttt{d}, \texttt{e}, \texttt{g}, \texttt{h}, \texttt{k}\}$$

The atom m has no head rule. The remaining support for b, f is a positive cycle with no initial fact. Neither is derived.

$$\boxed{X = \{\texttt{a}, \texttt{c}, \texttt{d}, \texttt{e}, \texttt{g}, \texttt{h}, \texttt{k}\}}$$

(b)

Rule numbers refer to the following order (one rule ends at each period).

```
conn(amsterdam,brussels).
conn(brussels,cologne).
conn(cologne,amsterdam).
tour(amsterdam,cologne) :- conn(amsterdam,brussels),
    conn(brussels,cologne).
tour(brussels,amsterdam) :- conn(brussels,cologne),
    conn(cologne,amsterdam).
tour(cologne,brussels) :- conn(amsterdam,brussels),
    conn(cologne,amsterdam).
finish(cologne) :- conn(amsterdam,amsterdam), tour(amsterdam,cologne).
finish(amsterdam) :- conn(brussels,amsterdam), tour(brussels,amsterdam).
finish(brussels) :- conn(cologne,amsterdam), tour(cologne,brussels).
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{conn(amsterdam,brussels)}\} \qquad \text{fact, rule 1}$$

$$M_2 = M_1 \cup \{\texttt{conn(brussels,cologne)}\} \qquad \text{fact, rule 2}$$

$$M_3 = M_2 \cup \{\texttt{conn(cologne,amsterdam)}\} \qquad \text{fact, rule 3}$$

$$M_4 = M_3 \cup \{\texttt{tour(amsterdam,cologne)}\} \qquad \text{rule 4}$$

$$M_5 = M_4 \cup \{\texttt{tour(brussels,amsterdam)}\} \qquad \text{rule 5}$$

$$M_6 = M_5 \cup \{\texttt{tour(cologne,brussels)}\} \qquad \text{rule 6}$$

$$M_7 = M_6 \cup \{\texttt{finish(brussels)}\} \qquad \text{rule 9}$$

No rule has a true body and a head outside M_7; hence this is the least fixed point.

$$\mathrm{Cn}(R) = \left\{ \begin{array}{c} \texttt{conn(amsterdam,brussels)},\ \texttt{conn(brussels,cologne)}, \\ \texttt{conn(cologne,amsterdam)},\ \texttt{finish(brussels)}, \\ \texttt{tour(amsterdam,cologne)}, \\ \texttt{tour(brussels,amsterdam)},\ \texttt{tour(cologne,brussels)} \end{array} \right\}$$

The bodies for `finish(cologne)` and `finish(amsterdam)` need absent connection atoms. Neither body is true.

$$\boxed{X = \left\{ \begin{array}{c} \texttt{conn(amsterdam,brussels)},\ \texttt{conn(brussels,cologne)}, \\ \texttt{conn(cologne,amsterdam)},\ \texttt{finish(brussels)}, \\ \texttt{tour(amsterdam,cologne)}, \\ \texttt{tour(brussels,amsterdam)},\ \texttt{tour(cologne,brussels)} \end{array} \right\}}$$

Exercise 1.2

Determine the reducts for the given candidate sets and decide which candidates are answer sets.

(a)

Rule numbers refer to the following order (one rule ends at each period).

```
a :- not d.
b :- not a.
b :- e, not d.
c :- a, not e.
c :- b, not e.
d :- not a, not e.
e :- a, d.
e :- b, not c.
```

Candidate X_1

$$X = \varnothing$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 1 is retained because $\{\texttt{d}\} \cap X = \varnothing$; its default negations are removed.

Rule 2 is retained because $\{\texttt{a}\} \cap X = \varnothing$; its default negations are removed.

Rule 3 is retained because $\{\texttt{d}\} \cap X = \varnothing$; its default negations are removed.

Rule 4 is retained because $\{\texttt{e}\} \cap X = \varnothing$; its default negations are removed.

Rule 5 is retained because $\{\texttt{e}\} \cap X = \varnothing$; its default negations are removed.

Rule 6 is retained because $\{\texttt{a},\ \texttt{e}\} \cap X = \varnothing$; its default negations are removed.

Rule 8 is retained because $\{\texttt{c}\} \cap X = \varnothing$; its default negations are removed.

Rule numbers refer to the following order (one rule ends at each period).

```
a.
b.
b :- e.
c :- a.
c :- b.
d.
e :- a, d.
e :- b.
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{a}\} \qquad \text{fact, rule 1}$$

$$M_2 = M_1 \cup \{\texttt{b}\} \qquad \text{fact, rule 2}$$

$$M_3 = M_2 \cup \{\texttt{d}\} \qquad \text{fact, rule 6}$$

$$M_4 = M_3 \cup \{\texttt{c}\} \qquad \text{rule 4}$$

$$M_5 = M_4 \cup \{\texttt{e}\} \qquad \text{rule 7}$$

No rule has a true body and a head outside M_5; hence this is the least fixed point.

$$\mathrm{Cn}(R) = \{\texttt{a},\ \texttt{b},\ \texttt{c},\ \texttt{d},\ \texttt{e}\}$$

$$\mathrm{Cn}(P^X) \neq X$$

$$\boxed{X \text{ is not an answer set.}}$$

Missing from the candidate: $\{\texttt{a},\ \texttt{b},\ \texttt{c},\ \texttt{d},\ \texttt{e}\}$.

Candidate X_2

$$X = \{\texttt{a},\ \texttt{c}\}$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 1 is retained because $\{\texttt{d}\} \cap X = \varnothing$; its default negations are removed.

Rule 2 is deleted because $\{\texttt{a}\} \subseteq X$.

Rule 3 is retained because $\{\texttt{d}\} \cap X = \varnothing$; its default negations are removed.

Rule 4 is retained because $\{\texttt{e}\} \cap X = \varnothing$; its default negations are removed.

Rule 5 is retained because $\{\texttt{e}\} \cap X = \varnothing$; its default negations are removed.

Rule 6 is deleted because $\{\texttt{a}\} \subseteq X$.

Rule 8 is deleted because $\{\texttt{c}\} \subseteq X$.

Rule numbers refer to the following order (one rule ends at each period).

```
a.
b :- e.
c :- a.
c :- b.
e :- a, d.
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{a}\} \qquad \text{fact, rule 1}$$

$$M_2 = M_1 \cup \{\texttt{c}\} \qquad \text{rule 3}$$

No rule has a true body and a head outside M_2; hence this is the least fixed point.

$$\mathrm{Cn}(R) = \{\texttt{a},\ \texttt{c}\}$$

$$\mathrm{Cn}(P^X) = X$$

$$\boxed{X \text{ is an answer set.}}$$

Candidate X_3

$$X = \{\texttt{a},\ \texttt{b},\ \texttt{e}\}$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 1 is retained because $\{\texttt{d}\} \cap X = \varnothing$; its default negations are removed.

Rule 2 is deleted because $\{\texttt{a}\} \subseteq X$.

Rule 3 is retained because $\{\texttt{d}\} \cap X = \varnothing$; its default negations are removed.

Rule 4 is deleted because $\{\texttt{e}\} \subseteq X$.

Rule 5 is deleted because $\{\texttt{e}\} \subseteq X$.

Rule 6 is deleted because $\{\texttt{a},\ \texttt{e}\} \subseteq X$.

Rule 8 is retained because $\{\texttt{c}\} \cap X = \varnothing$; its default negations are removed.

Rule numbers refer to the following order (one rule ends at each period).

```
a.
b :- e.
e :- a, d.
e :- b.
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{a}\} \qquad \text{fact, rule 1}$$

No rule has a true body and a head outside M_1; hence this is the least fixed point.

$$\mathrm{Cn}(R) = \{\texttt{a}\}$$

$$\mathrm{Cn}(P^X) \neq X$$

$$\boxed{X \text{ is not an answer set.}}$$

Unsupported candidate atoms: {b, e}.

(b)

Rule numbers refer to the following order (one rule ends at each period).

```
a :- not b.
b :- not c.
b :- not f.
c :- a.
c :- d.
d :- c, e.
d :- not b.
e :- not a, not f.
f :- not e.
```

Candidate X_1

$$X = \{\texttt{a}, \texttt{c}, \texttt{d}, \texttt{f}\}$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 1 is retained because $\{\texttt{b}\} \cap X = \varnothing$; its default negations are removed.

Rule 2 is deleted because $\{\texttt{c}\} \subseteq X$.

Rule 3 is deleted because $\{\texttt{f}\} \subseteq X$.

Rule 7 is retained because $\{\texttt{b}\} \cap X = \varnothing$; its default negations are removed.

Rule 8 is deleted because $\{\texttt{a}, \texttt{f}\} \subseteq X$.

Rule 9 is retained because $\{\texttt{e}\} \cap X = \varnothing$; its default negations are removed.

Rule numbers refer to the following order (one rule ends at each period).

```
a.
c :- a.
c :- d.
d :- c, e.
d.
f.
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{a}\} \qquad \text{fact, rule 1}$$

$$M_2 = M_1 \cup \{\texttt{d}\} \qquad \text{fact, rule 5}$$

$$M_3 = M_2 \cup \{\texttt{f}\} \qquad \text{fact, rule 6}$$

$$M_4 = M_3 \cup \{\texttt{c}\} \qquad \text{rule 2}$$

No rule has a true body and a head outside M_4; hence this is the least fixed point.

$$\mathrm{Cn}(R) = \{\texttt{a},\ \texttt{c},\ \texttt{d},\ \texttt{f}\}$$

$$\mathrm{Cn}(P^X) = X$$

$$\boxed{X \text{ is an answer set.}}$$

Candidate X_2

$$X = \{\texttt{b},\ \texttt{c},\ \texttt{d},\ \texttt{e}\}$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 1 is deleted because $\{\texttt{b}\} \subseteq X$.

Rule 2 is deleted because $\{\texttt{c}\} \subseteq X$.

Rule 3 is retained because $\{\texttt{f}\} \cap X = \varnothing$; its default negations are removed.

Rule 7 is deleted because $\{\texttt{b}\} \subseteq X$.

Rule 8 is retained because $\{\texttt{a},\ \texttt{f}\} \cap X = \varnothing$; its default negations are removed.

Rule 9 is deleted because $\{\texttt{e}\} \subseteq X$.

Rule numbers refer to the following order (one rule ends at each period).

```
b.
c :- a.
c :- d.
d :- c, e.
e.
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{b}\} \qquad \text{fact, rule 1}$$

$$M_2 = M_1 \cup \{\texttt{e}\} \qquad \text{fact, rule 5}$$

No rule has a true body and a head outside M_2; hence this is the least fixed point.

$$\mathrm{Cn}(R) = \{\texttt{b},\ \texttt{e}\}$$

$$\mathrm{Cn}(P^X) \neq X$$

$$\boxed{X \text{ is not an answer set.}}$$

Unsupported candidate atoms: $\{\texttt{c},\ \texttt{d}\}$.

Candidate X_3

$$X = \{\texttt{b},\ \texttt{e}\}$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 1 is deleted because $\{\texttt{b}\} \subseteq X$.

Rule 2 is retained because $\{\texttt{c}\} \cap X = \varnothing$; its default negations are removed.

Rule 3 is retained because $\{\texttt{f}\} \cap X = \varnothing$; its default negations are removed.

Rule 7 is deleted because $\{\texttt{b}\} \subseteq X$.

Rule 8 is retained because $\{\texttt{a},\ \texttt{f}\} \cap X = \varnothing$; its default negations are removed.

Rule 9 is deleted because $\{\texttt{e}\} \subseteq X$.

Rule numbers refer to the following order (one rule ends at each period).

```
b.
b.
c :- a.
c :- d.
d :- c, e.
e.
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{b}\} \qquad \text{fact, rule 1}$$

$$M_2 = M_1 \cup \{\texttt{e}\} \qquad \text{fact, rule 6}$$

No rule has a true body and a head outside M_2; hence this is the least fixed point.

$$\mathrm{Cn}(R) = \{\texttt{b},\ \texttt{e}\}$$

$$\mathrm{Cn}(P^X) = X$$

$$\boxed{X \text{ is an answer set.}}$$

Exercise 1.3

Find all answer sets of both programs from Exercise 1.2 without testing all interpretations.

(a)

If $a \in X$, then $d \notin X$: the only rule with head d is deleted by a. Suppose $e \in X$. The two rules for c are deleted. The rule b from *not* a is deleted, and d cannot be derived. The remaining support for b, e is the positive cycle b from e, e from b, with no initial fact. This contradicts stability. Hence $e \notin X$. Now b has no support and c follows from a. This gives $\{a, c\}$.

If $a \notin X$, stability requires $d \in X$, since otherwise the reduct would contain the fact a. Support for d requires $e \notin X$. The reduct contains the facts b, d and derives c from b. Neither rule for e can derive its head: a is absent and the rule with *not* c is deleted. This gives $\{b, c, d\}$. These two cases exhaust membership of a. □

$$\boxed{\#\,\mathrm{AS}(P) = 2}$$

Answer set 1 and reduct check

$$X = \{\texttt{a},\ \texttt{c}\}$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 1 is retained because $\{\texttt{d}\} \cap X = \varnothing$; its default negations are removed.

Rule 2 is deleted because $\{\texttt{a}\} \subseteq X$.

Rule 3 is retained because $\{\texttt{d}\} \cap X = \varnothing$; its default negations are removed.

Rule 4 is retained because $\{\texttt{e}\} \cap X = \varnothing$; its default negations are removed.

Rule 5 is retained because $\{\texttt{e}\} \cap X = \varnothing$; its default negations are removed.

Rule 6 is deleted because $\{\texttt{a}\} \subseteq X$.

Rule 8 is deleted because $\{\texttt{c}\} \subseteq X$.

Rule numbers refer to the following order (one rule ends at each period).

```
a.
b :- e.
c :- a.
c :- b.
e :- a, d.
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{a}\} \qquad \text{fact, rule 1}$$

$$M_2 = M_1 \cup \{\texttt{c}\} \qquad \text{rule 3}$$

No rule has a true body and a head outside M_2; hence this is the least fixed point.

$$\mathrm{Cn}(R) = \{\texttt{a},\ \texttt{c}\}$$

$$\mathrm{Cn}(P^X) = X$$

$$\boxed{X \text{ is an answer set.}}$$

Answer set 2 and reduct check

$$X = \{\texttt{b},\ \texttt{c},\ \texttt{d}\}$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 1 is deleted because $\{\texttt{d}\} \subseteq X$.

Rule 2 is retained because $\{\texttt{a}\} \cap X = \varnothing$; its default negations are removed.

Rule 3 is deleted because $\{\texttt{d}\} \subseteq X$.

Rule 4 is retained because $\{\texttt{e}\} \cap X = \varnothing$; its default negations are removed.

Rule 5 is retained because $\{\texttt{e}\} \cap X = \varnothing$; its default negations are removed.

Rule 6 is retained because $\{\texttt{a},\ \texttt{e}\} \cap X = \varnothing$; its default negations are removed.

Rule 8 is deleted because $\{\texttt{c}\} \subseteq X$.

Rule numbers refer to the following order (one rule ends at each period).

```
b.
c :- a.
c :- b.
d.
e :- a, d.
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{b}\} \qquad \text{fact, rule 1}$$

$$M_2 = M_1 \cup \{\texttt{d}\} \qquad \text{fact, rule 4}$$

$$M_3 = M_2 \cup \{\texttt{c}\} \qquad \text{rule 3}$$

No rule has a true body and a head outside M_3; hence this is the least fixed point.

$$\mathrm{Cn}(R) = \{\texttt{b},\ \texttt{c},\ \texttt{d}\}$$

$$\mathrm{Cn}(P^X) = X$$

$$\boxed{X \text{ is an answer set.}}$$

The complete list of answer sets is:

$$\boxed{X_1 = \{\texttt{a},\ \texttt{c}\}}$$

$$\boxed{X_2 = \{\texttt{b},\ \texttt{c},\ \texttt{d}\}}$$

(b)

If $e \in X$, the only rule for f is deleted, so $f \notin X$. Support for e requires $a \notin X$. The reduct has the fact b because f is absent. The rules with body *not* b are deleted. The rules for c, d form a positive cycle with no fact, so both atoms are absent. This gives $\{b, e\}$.

If $e \notin X$, the reduct contains the fact f. Split on b. For $b \notin X$, the reduct contains the facts a, d, and c follows from a. This gives $\{a, c, d, f\}$. For $b \in X$, a has no support. The rules for c, d cannot start a derivation, because e is absent and the fact-producing rule for d is deleted by b. Thus c, d are absent, and the reduct contains the fact b from *not* c. This gives $\{b, f\}$. The split on e and then b exhausts the cases. □

$$\boxed{\#\,\mathrm{AS}(P) = 3}$$

Answer set 1 and reduct check

$$X = \{\texttt{a},\ \texttt{c},\ \texttt{d},\ \texttt{f}\}$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 1 is retained because $\{\texttt{b}\} \cap X = \varnothing$; its default negations are removed.

Rule 2 is deleted because $\{\texttt{c}\} \subseteq X$.

Rule 3 is deleted because $\{\texttt{f}\} \subseteq X$.

Rule 7 is retained because $\{\texttt{b}\} \cap X = \varnothing$; its default negations are removed.

Rule 8 is deleted because $\{\texttt{a},\ \texttt{f}\} \subseteq X$.

Rule 9 is retained because $\{\texttt{e}\} \cap X = \varnothing$; its default negations are removed.

Rule numbers refer to the following order (one rule ends at each period).

```
a.
c :- a.
c :- d.
d :- c, e.
d.
f.
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{a}\} \qquad \text{fact, rule 1}$$

$$M_2 = M_1 \cup \{\texttt{d}\} \qquad \text{fact, rule 5}$$

$$M_3 = M_2 \cup \{\texttt{f}\} \qquad \text{fact, rule 6}$$

$$M_4 = M_3 \cup \{\texttt{c}\} \qquad \text{rule 2}$$

No rule has a true body and a head outside M_4; hence this is the least fixed point.

$$\mathrm{Cn}(R) = \{\texttt{a},\ \texttt{c},\ \texttt{d},\ \texttt{f}\}$$

$$\mathrm{Cn}(P^X) = X$$

$$\boxed{X \text{ is an answer set.}}$$

Answer set 2 and reduct check

$$X = \{\texttt{b},\ \texttt{e}\}$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 1 is deleted because $\{\texttt{b}\} \subseteq X$.

Rule 2 is retained because $\{\texttt{c}\} \cap X = \varnothing$; its default negations are removed.

Rule 3 is retained because $\{\texttt{f}\} \cap X = \varnothing$; its default negations are removed.

Rule 7 is deleted because $\{\texttt{b}\} \subseteq X$.

Rule 8 is retained because $\{\texttt{a},\ \texttt{f}\} \cap X = \varnothing$; its default negations are removed.

Rule 9 is deleted because $\{\texttt{e}\} \subseteq X$.

Rule numbers refer to the following order (one rule ends at each period).

```
b.
b.
c :- a.
c :- d.
d :- c, e.
e.
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{b}\} \qquad \text{fact, rule 1}$$

$$M_2 = M_1 \cup \{\texttt{e}\} \qquad \text{fact, rule 6}$$

No rule has a true body and a head outside M_2; hence this is the least fixed point.

$$\mathrm{Cn}(R) = \{\texttt{b},\ \texttt{e}\}$$

$$\mathrm{Cn}(P^X) = X$$

$$\boxed{X \text{ is an answer set.}}$$

Answer set 3 and reduct check

$$X = \{\texttt{b},\ \texttt{f}\}$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 1 is deleted because $\{\texttt{b}\} \subseteq X$.

Rule 2 is retained because $\{\texttt{c}\} \cap X = \varnothing$; its default negations are removed.

Rule 3 is deleted because $\{\texttt{f}\} \subseteq X$.

Rule 7 is deleted because $\{\texttt{b}\} \subseteq X$.

Rule 8 is deleted because $\{\texttt{f}\} \subseteq X$.

Rule 9 is retained because $\{\texttt{e}\} \cap X = \varnothing$; its default negations are removed.

Rule numbers refer to the following order (one rule ends at each period).

```
b.
c :- a.
c :- d.
d :- c, e.
f.
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{b}\} \qquad \text{fact, rule 1}$$

$$M_2 = M_1 \cup \{\texttt{f}\} \qquad \text{fact, rule 5}$$

No rule has a true body and a head outside M_2; hence this is the least fixed point.

$$\mathrm{Cn}(R) = \{\texttt{b},\ \texttt{f}\}$$

$$\mathrm{Cn}(P^X) = X$$

$$\boxed{X \text{ is an answer set.}}$$

The complete list of answer sets is:

$$\boxed{X_1 = \{\texttt{a},\ \texttt{c},\ \texttt{d},\ \texttt{f}\}}$$

$$\boxed{X_2 = \{\texttt{b},\ \texttt{e}\}}$$

$$\boxed{X_3 = \{\texttt{b},\ \texttt{f}\}}$$

Exercise 1.4

Determine all answer sets of the following normal logic programs.

(a)

Rule numbers refer to the following order (one rule ends at each period).

```
canary(tweety).
penguin(tux).
bird(tweety) :- canary(tweety).
bird(tux) :- canary(tux).
bird(tweety) :- penguin(tweety).
```

```
bird(tux) :- penguin(tux).
walks(tweety) :- penguin(tweety).
walks(tux) :- penguin(tux).
walks(tweety) :- bird(tweety), not flies(tweety).
walks(tux) :- bird(tux), not flies(tux).
flies(tweety) :- bird(tweety), not walks(tweety).
flies(tux) :- bird(tux), not walks(tux).
```

The facts derive both bird atoms and `walks(tux)`. Therefore the rule for `flies(tux)` is deleted. For Tweety, neither walking nor flying would make both reduct rules fire; both walking and flying would delete their two supporting rules. Hence exactly one holds. The two candidates below exhaust this choice; Tux always walks and never flies. □

$$\boxed{\#\,\mathrm{AS}(P) = 2}$$

Answer set 1 and reduct check

$$X = \left\{ \begin{array}{l} \texttt{bird(tux)},\ \texttt{bird(tweety)},\ \texttt{canary(tweety)}, \\ \texttt{flies(tweety)},\ \texttt{penguin(tux)},\ \texttt{walks(tux)} \end{array} \right\}$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 9 is deleted because $\{\texttt{flies(tweety)}\} \subseteq X$.

Rule 10 is retained because $\{\texttt{flies(tux)}\} \cap X = \varnothing$; its default negations are removed.

Rule 11 is retained because $\{\texttt{walks(tweety)}\} \cap X = \varnothing$; its default negations are removed.

Rule 12 is deleted because $\{\texttt{walks(tux)}\} \subseteq X$.

Rule numbers refer to the following order (one rule ends at each period).

```
canary(tweety).
penguin(tux).
bird(tweety) :- canary(tweety).
bird(tux) :- canary(tux).
bird(tweety) :- penguin(tweety).
bird(tux) :- penguin(tux).
walks(tweety) :- penguin(tweety).
walks(tux) :- penguin(tux).
walks(tux) :- bird(tux).
flies(tweety) :- bird(tweety).
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{canary(tweety)}\} \qquad \text{fact, rule 1}$$

$$M_2 = M_1 \cup \{\texttt{penguin(tux)}\} \qquad \text{fact, rule 2}$$

$$M_3 = M_2 \cup \{\texttt{bird(tux)}\} \qquad \text{rule 6}$$

$$M_4 = M_3 \cup \{\texttt{bird(tweety)}\} \qquad \text{rule 3}$$

$$M_5 = M_4 \cup \{\texttt{walks(tux)}\} \qquad \text{rule 8}$$

$$M_6 = M_5 \cup \{\texttt{flies(tweety)}\} \qquad \text{rule 10}$$

No rule has a true body and a head outside M_6; hence this is the least fixed point.

$$\mathrm{Cn}(R) = \left\{ \begin{matrix} \texttt{bird(tux)},\ \texttt{bird(tweety)},\ \texttt{canary(tweety)}, \\ \texttt{flies(tweety)},\ \texttt{penguin(tux)},\ \texttt{walks(tux)} \end{matrix} \right\}$$

$$\mathrm{Cn}(P^X) = X$$

$$\boxed{X \text{ is an answer set.}}$$

Answer set 2 and reduct check

$$X = \left\{ \begin{matrix} \texttt{bird(tux)},\ \texttt{bird(tweety)},\ \texttt{canary(tweety)}, \\ \texttt{penguin(tux)},\ \texttt{walks(tux)},\ \texttt{walks(tweety)} \end{matrix} \right\}$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 9 is retained because $\{\texttt{flies(tweety)}\} \cap X = \varnothing$; its default negations are removed.

Rule 10 is retained because $\{\texttt{flies(tux)}\} \cap X = \varnothing$; its default negations are removed.

Rule 11 is deleted because $\{\texttt{walks(tweety)}\} \subseteq X$.

Rule 12 is deleted because $\{\texttt{walks(tux)}\} \subseteq X$.

Rule numbers refer to the following order (one rule ends at each period).

```
canary(tweety).
penguin(tux).
bird(tweety) :- canary(tweety).
bird(tux) :- canary(tux).
bird(tweety) :- penguin(tweety).
bird(tux) :- penguin(tux).
walks(tweety) :- penguin(tweety).
walks(tux) :- penguin(tux).
walks(tweety) :- bird(tweety).
walks(tux) :- bird(tux).
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{canary(tweety)}\} \qquad \text{fact, rule 1}$$

$$M_2 = M_1 \cup \{\texttt{penguin(tux)}\} \qquad \text{fact, rule 2}$$

$$M_3 = M_2 \cup \{\texttt{bird(tux)}\} \qquad \text{rule 6}$$

$$M_4 = M_3 \cup \{\texttt{bird(tweety)}\} \qquad \text{rule 3}$$

$$M_5 = M_4 \cup \{\texttt{walks(tux)}\} \qquad \text{rule 8}$$

$$M_6 = M_5 \cup \{\texttt{walks(tweety)}\} \qquad \text{rule 9}$$

No rule has a true body and a head outside M_6; hence this is the least fixed point.

$$\mathrm{Cn}(R) = \left\{\begin{array}{c}\texttt{bird(tux)},\ \texttt{bird(tweety)},\ \texttt{canary(tweety)},\\ \texttt{penguin(tux)},\ \texttt{walks(tux)},\ \texttt{walks(tweety)}\end{array}\right\}$$

$$\mathrm{Cn}(P^X) = X$$

$$\boxed{X \text{ is an answer set.}}$$

The complete list of answer sets is:

$$\boxed{X_1 = \left\{\begin{array}{c}\texttt{bird(tux)},\ \texttt{bird(tweety)},\ \texttt{canary(tweety)},\\ \texttt{flies(tweety)},\ \texttt{penguin(tux)},\ \texttt{walks(tux)}\end{array}\right\}}$$

$$\boxed{X_2 = \left\{\begin{array}{c}\texttt{bird(tux)},\ \texttt{bird(tweety)},\ \texttt{canary(tweety)},\\ \texttt{penguin(tux)},\ \texttt{walks(tux)},\ \texttt{walks(tweety)}\end{array}\right\}}$$

(b)

Rule numbers refer to the following order (one rule ends at each period).

```
num(1).
num(2).
num(3).
lucky(1) :- num(1), not unlucky(1).
lucky(2) :- num(2), not unlucky(2).
lucky(3) :- num(3), not unlucky(3).
unlucky(1) :- num(1), not lucky(1).
unlucky(2) :- num(2), not lucky(2).
unlucky(3) :- num(3), not lucky(3).
p :- lucky(1), lucky(2), not p.
p :- lucky(1), lucky(3), not p.
p :- lucky(2), lucky(3), not p.
```

For each number, exactly one of `lucky` and `unlucky` holds: neither would derive both in the reduct, and both would delete both supporting rules. If $p \in X$, all its rules are deleted, so p has no reduct support. Thus $p \notin X$. With p absent, each pair of lucky atoms would derive p. At most one number can be lucky. With no lucky number, all three are unlucky. With lucky number 1, numbers 2 and 3 are unlucky. With lucky number 2, numbers 1 and 3 are unlucky. With lucky number 3, numbers 1 and 2 are unlucky. These four cases exhaust the possibilities. □

$$\boxed{\#\,\mathrm{AS}(P) = 4}$$

Answer set 1 and reduct check

$$X = \left\{\begin{array}{c}\texttt{lucky(1)},\ \texttt{num(1)},\ \texttt{num(2)},\ \texttt{num(3)},\ \texttt{unlucky(2)},\\ \texttt{unlucky(3)}\end{array}\right\}$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 4 is retained because $\{\texttt{unlucky(1)}\} \cap X = \varnothing$; its default negations are removed.

Rule 5 is deleted because $\{\texttt{unlucky(2)}\} \subseteq X$.

Rule 6 is deleted because $\{\texttt{unlucky(3)}\} \subseteq X$.

Rule 7 is deleted because $\{\texttt{lucky(1)}\} \subseteq X$.

Rule 8 is retained because $\{\texttt{lucky(2)}\} \cap X = \varnothing$; its default negations are removed.

Rule 9 is retained because $\{\texttt{lucky(3)}\} \cap X = \varnothing$; its default negations are removed.

Rule 10 is retained because $\{\texttt{p}\} \cap X = \varnothing$; its default negations are removed.

Rule 11 is retained because $\{\texttt{p}\} \cap X = \varnothing$; its default negations are removed.

Rule 12 is retained because $\{\texttt{p}\} \cap X = \varnothing$; its default negations are removed.

Rule numbers refer to the following order (one rule ends at each period).

```
num(1).
num(2).
num(3).
lucky(1) :- num(1).
unlucky(2) :- num(2).
unlucky(3) :- num(3).
p :- lucky(1), lucky(2).
p :- lucky(1), lucky(3).
p :- lucky(2), lucky(3).
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{num(1)}\} \qquad \text{fact, rule 1}$$

$$M_2 = M_1 \cup \{\texttt{num(2)}\} \qquad \text{fact, rule 2}$$

$$M_3 = M_2 \cup \{\texttt{num(3)}\} \qquad \text{fact, rule 3}$$

$$M_4 = M_3 \cup \{\texttt{lucky(1)}\} \qquad \text{rule 4}$$

$$M_5 = M_4 \cup \{\texttt{unlucky(2)}\} \qquad \text{rule 5}$$

$$M_6 = M_5 \cup \{\texttt{unlucky(3)}\} \qquad \text{rule 6}$$

No rule has a true body and a head outside M_6; hence this is the least fixed point.

$$\mathrm{Cn}(R) = \left\{\begin{matrix}\texttt{lucky(1)},\ \texttt{num(1)},\ \texttt{num(2)},\ \texttt{num(3)},\ \texttt{unlucky(2)},\\ \texttt{unlucky(3)}\end{matrix}\right\}$$

$$\mathrm{Cn}(P^X) = X$$

$$\boxed{X \text{ is an answer set.}}$$

Answer set 2 and reduct check

$$X = \left\{\begin{matrix}\texttt{lucky(2)},\ \texttt{num(1)},\ \texttt{num(2)},\ \texttt{num(3)},\ \texttt{unlucky(1)},\\ \texttt{unlucky(3)}\end{matrix}\right\}$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 4 is deleted because $\{\texttt{unlucky(1)}\} \subseteq X$.

Rule 5 is retained because $\{\texttt{unlucky(2)}\} \cap X = \varnothing$; its default negations are removed.

Rule 6 is deleted because $\{\texttt{unlucky(3)}\} \subseteq X$.

Rule 7 is retained because $\{\texttt{lucky(1)}\} \cap X = \varnothing$; its default negations are removed.

Rule 8 is deleted because $\{\texttt{lucky(2)}\} \subseteq X$.

Rule 9 is retained because $\{\texttt{lucky(3)}\} \cap X = \varnothing$; its default negations are removed.

Rule 10 is retained because $\{\texttt{p}\} \cap X = \varnothing$; its default negations are removed.

Rule 11 is retained because $\{\texttt{p}\} \cap X = \varnothing$; its default negations are removed.

Rule 12 is retained because $\{\texttt{p}\} \cap X = \varnothing$; its default negations are removed.

Rule numbers refer to the following order (one rule ends at each period).

```
num(1).
num(2).
num(3).
lucky(2) :- num(2).
unlucky(1) :- num(1).
unlucky(3) :- num(3).
p :- lucky(1), lucky(2).
p :- lucky(1), lucky(3).
p :- lucky(2), lucky(3).
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{num(1)}\} \qquad \text{fact, rule 1}$$

$$M_2 = M_1 \cup \{\texttt{num(2)}\} \qquad \text{fact, rule 2}$$

$$M_3 = M_2 \cup \{\texttt{num(3)}\} \qquad \text{fact, rule 3}$$

$$M_4 = M_3 \cup \{\texttt{lucky(2)}\} \qquad \text{rule 4}$$

$$M_5 = M_4 \cup \{\texttt{unlucky(1)}\} \qquad \text{rule 5}$$

$$M_6 = M_5 \cup \{\texttt{unlucky(3)}\} \qquad \text{rule 6}$$

No rule has a true body and a head outside M_6; hence this is the least fixed point.

$$\mathrm{Cn}(R) = \left\{ \begin{array}{c} \texttt{lucky(2)},\ \texttt{num(1)},\ \texttt{num(2)},\ \texttt{num(3)},\ \texttt{unlucky(1)}, \\ \texttt{unlucky(3)} \end{array} \right\}$$

$$\mathrm{Cn}(P^X) = X$$

$$\boxed{X \text{ is an answer set.}}$$

Answer set 3 and reduct check

$$X = \left\{ \begin{array}{c} \texttt{lucky(3)},\ \texttt{num(1)},\ \texttt{num(2)},\ \texttt{num(3)},\ \texttt{unlucky(1)}, \\ \texttt{unlucky(2)} \end{array} \right\}$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 4 is deleted because $\{\texttt{unlucky(1)}\} \subseteq X$.

Rule 5 is deleted because $\{\texttt{unlucky(2)}\} \subseteq X$.

Rule 6 is retained because $\{\texttt{unlucky(3)}\} \cap X = \varnothing$; its default negations are removed.

Rule 7 is retained because $\{\texttt{lucky(1)}\} \cap X = \varnothing$; its default negations are removed.

Rule 8 is retained because $\{\texttt{lucky(2)}\} \cap X = \varnothing$; its default negations are removed.

Rule 9 is deleted because $\{\texttt{lucky(3)}\} \subseteq X$.

Rule 10 is retained because $\{\texttt{p}\} \cap X = \varnothing$; its default negations are removed.

Rule 11 is retained because $\{\texttt{p}\} \cap X = \varnothing$; its default negations are removed.

Rule 12 is retained because $\{\texttt{p}\} \cap X = \varnothing$; its default negations are removed.

Rule numbers refer to the following order (one rule ends at each period).

```
num(1).
num(2).
num(3).
lucky(3) :- num(3).
unlucky(1) :- num(1).
unlucky(2) :- num(2).
p :- lucky(1), lucky(2).
p :- lucky(1), lucky(3).
p :- lucky(2), lucky(3).
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{num(1)}\} \qquad \text{fact, rule 1}$$

$$M_2 = M_1 \cup \{\texttt{num(2)}\} \qquad \text{fact, rule 2}$$

$$M_3 = M_2 \cup \{\texttt{num(3)}\} \qquad \text{fact, rule 3}$$

$$M_4 = M_3 \cup \{\texttt{lucky(3)}\} \qquad \text{rule 4}$$

$$M_5 = M_4 \cup \{\texttt{unlucky(1)}\} \qquad \text{rule 5}$$

$$M_6 = M_5 \cup \{\texttt{unlucky(2)}\} \qquad \text{rule 6}$$

No rule has a true body and a head outside M_6; hence this is the least fixed point.

$$\mathrm{Cn}(R) = \left\{ \begin{array}{c} \texttt{lucky(3)},\ \texttt{num(1)},\ \texttt{num(2)},\ \texttt{num(3)},\ \texttt{unlucky(1)}, \\ \texttt{unlucky(2)} \end{array} \right\}$$

$$\mathrm{Cn}(P^X) = X$$

$\boxed{X \text{ is an answer set.}}$

Answer set 4 and reduct check

$$X = \left\{ \begin{array}{c} \texttt{num(1)},\ \texttt{num(2)},\ \texttt{num(3)},\ \texttt{unlucky(1)},\ \texttt{unlucky(2)}, \\ \texttt{unlucky(3)} \end{array} \right\}$$

Reduct $R = P^X$: delete a rule if a default-negated atom belongs to X; remove the remaining default negations. Deletion decisions use the order in P; the subsequent derivation uses the order in the displayed reduct R. Rule 4 is deleted because $\{\texttt{unlucky(1)}\} \subseteq X$.

Rule 5 is deleted because $\{\texttt{unlucky(2)}\} \subseteq X$.

Rule 6 is deleted because $\{\texttt{unlucky(3)}\} \subseteq X$.

Rule 7 is retained because $\{\texttt{lucky(1)}\} \cap X = \varnothing$; its default negations are removed.

Rule 8 is retained because $\{\texttt{lucky(2)}\} \cap X = \varnothing$; its default negations are removed.

Rule 9 is retained because $\{\texttt{lucky(3)}\} \cap X = \varnothing$; its default negations are removed.

Rule 10 is retained because $\{\texttt{p}\} \cap X = \varnothing$; its default negations are removed.

Rule 11 is retained because $\{\texttt{p}\} \cap X = \varnothing$; its default negations are removed.

Rule 12 is retained because $\{\texttt{p}\} \cap X = \varnothing$; its default negations are removed.

Rule numbers refer to the following order (one rule ends at each period).

```
num(1).
num(2).
num(3).
unlucky(1) :- num(1).
unlucky(2) :- num(2).
unlucky(3) :- num(3).
p :- lucky(1), lucky(2).
p :- lucky(1), lucky(3).
p :- lucky(2), lucky(3).
```

$$M_0 = \varnothing$$

$$M_1 = M_0 \cup \{\texttt{num(1)}\} \qquad \text{fact, rule 1}$$

$$M_2 = M_1 \cup \{\texttt{num(2)}\} \qquad \text{fact, rule 2}$$

$$M_3 = M_2 \cup \{\texttt{num(3)}\} \qquad \text{fact, rule 3}$$

$$M_4 = M_3 \cup \{\texttt{unlucky(1)}\} \qquad \text{rule 4}$$

$$M_5 = M_4 \cup \{\texttt{unlucky(2)}\} \qquad \text{rule 5}$$

$$M_6 = M_5 \cup \{\texttt{unlucky(3)}\} \qquad \text{rule 6}$$

No rule has a true body and a head outside M_6; hence this is the least fixed point.

$$\mathrm{Cn}(R) = \left\{ \begin{array}{c} \texttt{num(1)},\ \texttt{num(2)},\ \texttt{num(3)},\ \texttt{unlucky(1)},\ \texttt{unlucky(2)}, \\ \texttt{unlucky(3)} \end{array} \right\}$$

$$\mathrm{Cn}(P^X) = X$$

X is an answer set.

The complete list of answer sets is:

$$X_1 = \left\{ \begin{array}{c} \texttt{lucky(1)}, \texttt{num(1)}, \texttt{num(2)}, \texttt{num(3)}, \texttt{unlucky(2)}, \\ \texttt{unlucky(3)} \end{array} \right\}$$

$$X_2 = \left\{ \begin{array}{c} \texttt{lucky(2)}, \texttt{num(1)}, \texttt{num(2)}, \texttt{num(3)}, \texttt{unlucky(1)}, \\ \texttt{unlucky(3)} \end{array} \right\}$$

$$X_3 = \left\{ \begin{array}{c} \texttt{lucky(3)}, \texttt{num(1)}, \texttt{num(2)}, \texttt{num(3)}, \texttt{unlucky(1)}, \\ \texttt{unlucky(2)} \end{array} \right\}$$

$$X_4 = \left\{ \begin{array}{c} \texttt{num(1)}, \texttt{num(2)}, \texttt{num(3)}, \texttt{unlucky(1)}, \texttt{unlucky(2)}, \\ \texttt{unlucky(3)} \end{array} \right\}$$